

Quantitative Coding of Negotiation Behavior

LAURIE R. WEINGART*

David A. Tepper School of Business Carnegie Mellon University, 5000 Forbes Avenue, Pittsburgh, PA 15213 USA E-mail: weingart@cmu.edu

MARA OLEKALNS**

Melbourne Business School, University of Melbourne, 200 Leicester Street Carlton VIC 3053 Australia E-mail: m.olekalns@mbs.edu

PHILIP L. SMITH***

Department of Psychology, University of Melbourne, Victoria, 3010 Australia E-mail: philip@unimelb.edu.au

Abstract. The examination of negotiation processes is seen by many researchers as an insurmountable task largely because the required methods are unfamiliar and labor-intensive. In this article, we shed light on a fundamental step in studying negotiation processes, the quantitative

* Laurie R. Weingart is a Professor of Organizational Behavior at the David A. Tepper School of Business, Carnegie Mellon University. She holds a Ph.D. in Organizational Behavior from the J.L. Kellogg Graduate School of Management at Northwestern University. Her research focuses on conflict management in work groups and the tactical behavior and cognitive processes of negotiators in both dyads and groups. Prof. Weingart publishes in top-tier management and social psychology journals, has served in leadership roles in both the Academy of Management and the International Association for Conflict Management, and serves on the editorial boards of *Academy of Management Review* and *Organizational Behavior and Human Decision Processes*.

** Mara Olekalns is a Professor of Management (Negotiations) at the Melbourne Business School, University of Melbourne. She holds a PhD in Psychology from the University of Adelaide. Her research focuses on communication processes in negotiation. Her research investigates how strategy sequences shape negotiation outcomes. Her work on communication processes in negotiation has been published in *Human Communication Research*, *Journal of Experimental Social Psychology*, *Personality and Social Psychology Bulletin*, and *Organizational Behavior and Human Decision Processes*.

*** Philip L. Smith is a Professor in the Department of Psychology, University of Melbourne. He holds a PhD in Psychology from the University of Adelaide. His primary research interest is in building quantitative models of the human visual system. This expertise is extended to modeling the relationship between strategy sequences and negotiation outcomes. His research has been published in *Psychological Review*, *Human Communication Research*, *Journal of Experimental Social Psychology*, *Personality and Social Psychology Bulletin*, and *Organizational Behavior and Human Decision Processes*.

coding of data. Relying on videotapes as the primary source of data, we review the steps required to extract usable quantitative data and the lessons we've learned in doing so in our own research. We review our experience working with one large negotiation dataset, Towers Market II, to illustrate two steps within the larger research process: developing a coding scheme and coding the data. We then go on to discuss some of the issues that need to be resolved before data analysis begins.

Keywords: Negotiation, communication, research methods, coding, dynamic processes

We have spent most of our academic careers studying negotiation processes directly. Why? Because we are crazy? Because no one else will do it? Perhaps, but we like to think we study negotiation processes because the approach provides unique insights into how negotiations unfold. Direct examination of negotiation processes provides information about what negotiators actually do, rather than what they planned to do or what they thought they did. The resulting data can be used to capture general strategies employed by negotiators (through frequency analysis), how they employed those strategies (through sequential analysis), and when they did so (through phase analysis). Besides gaining a better understanding of the negotiation process itself, this information can be used to predict whether negotiators reach agreement as well as the quality of their agreements, and to determine how negotiators can be better trained to effectively negotiate.

Much of our research has been carried out using an experimental paradigm designed to test the relationships between the negotiation context, negotiation behaviors, and outcomes. We videotape dyads and groups negotiating in role-play exercises. The videotapes then become our primary source of data. In this article we review the steps that extract usable data from these videotapes and the lessons we have learned along the way. In this article, which builds on previous discussions of studying group (Weingart, 1997) and negotiation (Brett, Weingart, and Olekalns 2004) processes, we review our experience working with one large negotiation dataset, Towers Market II, to illustrate two steps within the larger research process: developing a coding scheme and coding the data. We then go on to discuss some of the issues that need to be resolved before data analysis begins.

Coding Scheme Development

The first step in studying negotiation processes is to determine what types of behavior are theoretically relevant. This step is not to be taken lightly, as it has a profound influence on one's ability to test and support hypotheses.

For example, in our early research, we somewhat surprisingly found little or no relationship between information-sharing and quality of agreements (Weingart, Thompson, Bazerman, and Carroll 1990). We realized that this may have occurred because we had not distinguished between different types of information. Subsequent studies proved us correct: Priority information exchange positively influenced the quality of agreements whereas information about preferences and positions negatively (or neutrally) influenced them (Hyder, Prietula, and Weingart 2000; Olekalns & Smith 2000; Olekalns, Smith and Walsh 1996; Weingart, Hyder, and Prietula 1996).

In this section, we consider the choices a researcher interested in negotiation processes must make in developing a coding scheme. In making these choices, researchers need to consider three questions: Should the scheme focus on task or relational aspects of process? Should it be theory – or data-driven? What steps should we take to ensure reliability? For more general discussions regarding how to construct a coding scheme see Folger, Hewes, and Poole (1984), Hewes (1979), Poole, Folger, and Hewes (1987) and Trujillo (1986).

Level of coding

When coding negotiations, we can focus on two aspects of behavior: their impact on outcomes and their impact on the underlying relationship. These aspects of behavior are not mutually exclusive. Rather, the issue is one of relative emphasis. Whereas coding schemes that evolved from negotiation theory emphasize outcomes, those that evolved from communication theory emphasize the relational aspects of strategy choice. Ultimately, there is considerable overlap between the kinds of tactics that researchers working from these two perspectives code for. The approaches differ in that coding schemes based on negotiation theory have been developed bottom up, with functions inferred from tactics, whereas schemes developed based on communication theory have been developed top-down, with functions driving the identification of tactics. The more important decision, for researchers, is which level of analysis best suits their needs and how extensive their list of tactics should be.

Coding schemes stemming from negotiation theory have their origins in work by Pruitt and colleagues in the 1970s and 1980s (e.g., Carnevale, Pruitt, and Seilheimer 1981; Pruitt 1981; Pruitt & Lewis 1975) and Bales' work in the 1950s (Bales 1950). These schemes focus on the substantive aspects of negotiation: value claiming and value creation. Such schemes code for specific tactics such as threats, offers, questions regarding preferences and priorities, creative solutions, and information seeking. These specific behaviors can,

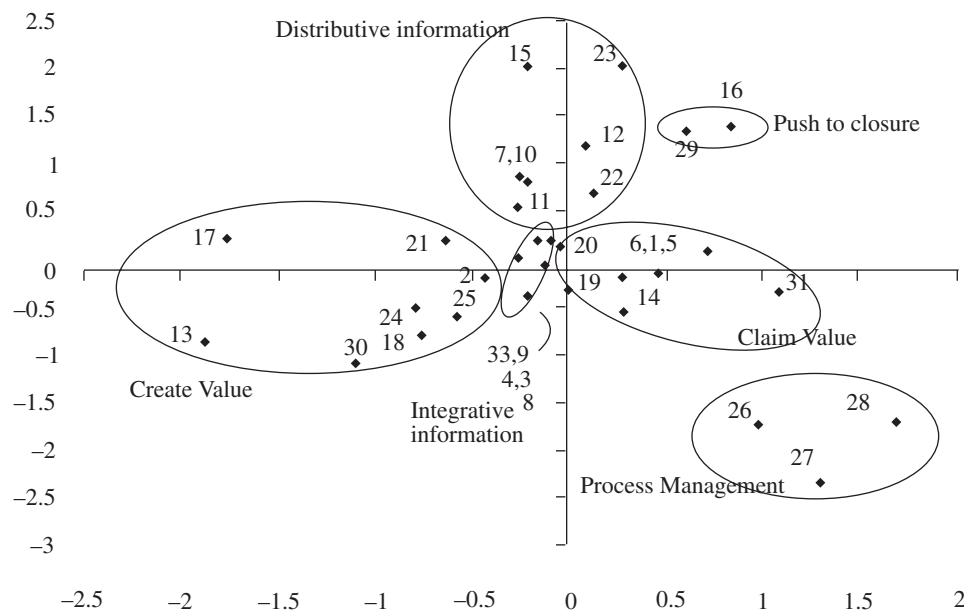
however, be linked to one of two strategic orientations – integrative or distributive – and so convey implicit information about the underlying relationship. The scheme that we developed for our Towers Market research is an example of this type of coding scheme (see Table 1 for the list of coding categories).

Table 1. Towers II Negotiation Behaviors in Strategy Clusters

Strategy	Behaviors included in strategy cluster	Category Number ^a
Integrative Information	States issue preferences	3
	States issue priorities	4
	Asks questions about preferences	8
	Asks questions about priorities	9
	Off-task comments (relationship building)	33
Create Value	Makes multi-issue offer	2
	Shows insight	13
	Notes general differences	17
	Notes general similarities	18
	Makes positive comments	21
	Suggests compromise	24
	Suggests package trade-offs	25
	Other process suggestions	30
Distributive Information	Facts	7
	Asks for bottom line	10
	Asks about others' substantiation (attack arguments)	11
	Asks miscellaneous task related questions	12
	Notes differences	15
	Negative reactions	22
	Suggests discuss one issue	23
Claim Value	Makes single issue offer	1
	Refers to bottom line	5
	Substantiates Position	6
	Refers to mutual interests to influence other party	14
	Threats	19
	Refers to power	20
	Suggests creative solutions to meet own interests	31
Push to closure	Time checks	29
	Notes similarities	16
Process management	Suggests using reciprocity	26
	Suggests vote	27
	Suggests to move on	28

^a Numbers correspond to those in Figure 1.

Figure 1. Towers II Correspondence Analysis Results



^a Numbers correspond to those in Table 1.

Communication theory based schemes start by considering the relational functions served by communication: either to highlight similarities and bring people together *or* to highlight differences and to push people apart. In the context of negotiations, this has led to the identification of three categories of negotiation behaviors, integrating, attacking and defending, and a set of specific tactics that fit within these categories (e.g., Donohue, Diez and Hamilton 1984; Putnam & Jones 1982). Unlike early schemes derived from negotiation theory, those derived from communication theory code not just for substantive tactics but also for relational tactics (e.g., showing support for the other party). In addition, communication researchers hypothesize that communication can serve two functions: one is to provide a response to the other party; the other is to cue the other party's subsequent behavior. Thus, it is possible to code any speaking turn twice, thereby capturing both aspects of communication (see, for example, Donohue et al. 1984; Olekalns & Smith 2000). Such an approach is especially useful when speaking turns are long and incorporate several different tactics.

Developing and refining the coding scheme

Coding schemes can be either theoretically derived (e.g., Weingart et al., 1996) or developed using a more grounded approach, derived from observation of negotiations themselves (refer to Putnam's chapter in same volume). However, the distinction between traditional theory-driven approaches and more data-driven approaches is necessarily blurred in practice. Human behavior is typically too complex for a researcher to be able to anticipate all relevant behaviors without some direct experience of how people interact. Coding schemes and their associated manuals should be considered living documents, open to revision and clarification during the development process, but then locked in when coding begins.

We have used a hybrid strategy of coding scheme development in order to capitalize on the benefits of both approaches. For example, when we initially developed the Towers Market coding scheme (Weingart, Bennett, and Brett 1993) we used negotiation theories and textbooks as well as existing coding schemes to identify theoretically important negotiation behaviors (e.g., Lewicki, Saunders, and Minton 1985; Pruitt 1981; Pruitt & Carnevale 1993; Walton & McKersie 1965). These yielded a set of task-specific behaviors such as information exchange, offers, tradeoffs across issues, and argumentation. Because we were also interested in how group members managed the process, we added several different types of process statements. After developing the categories, we tested them out on some sample transcripts. Some categories worked, others did not, so we altered the coding scheme accordingly.

The next step involves refining the coding scheme. This is best done by having your coders independently code sections of transcripts and then checking for ease of usage and agreement. This will help reveal categories that are too broad (when types of behavior that seem quite different are similarly classified), and others that are too narrow or too similar to other categories (when deciding between two categories for a given behavior is repeatedly a problem). Relevant questions include: Can coders differentiate between the categories when applying them? Are there categories for all relevant behaviors? Can the scheme be applied reliably (discussed below)? If not, is it because of problems in training the coders or in the design of the scheme?

Coding data

In parallel with developing this coding scheme we developed a coding manual, that is, a set of application rules/guidelines for the coders, including identification of the unit to be coded (thought, sentence, speaking turn), category labels, definitions, and rules of thumb for distinguishing between cate-

gories and using context (i.e., statements surrounding the unit of interest) to interpret meaning. Our coding manuals typically include general rules about unitizing and coding as well as definitions of content categories, examples, and exceptions to the rules. This manual is used to train coders and to identify problems in applying the coding scheme. In particular, if disagreements about the application of a particular code appear to be haphazard and difficult to resolve, then the problem may reside in the design of the coding scheme itself. Categories might overlap or be missing. These definitional issues need to be resolved before coding proper begins. There are two discrete steps in coding process data: 1) unitizing the data, or identifying the unit of analysis to be coded, and 2) content coding the data, or applying labels to units as a means of capturing what occurred. Next, we discuss issues of unitizing and unitizing reliability, followed by a discussion of content coding reliability and validity.

Unitizing

Coders must first identify the unit of analysis for coding. This can vary from a single utterance to an interaction between two individuals. Speaking turns, that is, units that include all actions and/or statements made by an individual while he or she holds the floor, are often used when studying negotiation because they allow us to examine how one negotiator responds to another. Speaking turns can be subdivided into *acts*, that is, identifiable ideas or thoughts, which can be used when interested in the content of the dialogue said rather than the interactive nature of the negotiation. In the Towers II data, we unitized at the level of acts, that is, we coded each identifiable idea, but also retained information about speaking turns to maximize our flexibility. Note that one speaking turn can be made up of several thought units.

Decisions regarding choice of unit type must be linked to the research question being asked, with special emphasis on the appropriate level of analysis and to the question of where in action and speech relevant meaning resides. Meaning can be lost if we select too small a unit, because the individual statements convey a different meaning to that conveyed by a speaking turn. We can also add redundancy as a result of separating immediate restatements, that which simply repeat previous tactics rather than adding new information. In contrast, by selecting too large a unit, information can be lost. **If multiple categories of statements are made during a speaking turn, the researcher must decide which code best represents the behavior within a given unit.** Dominance schemes, which identify the kinds of behavior that are expected to have the greatest impact on the interaction, can be developed to provide rules of thumb for these decisions (e.g., Weingart et al. 1993). Alternatively,

the first or the last code within the unit might be retained. Regardless of the approach that is used, the risk of losing valuable information remains.

Coders must meet two criteria of reliability: *unitizing reliability* and *interpretive reliability*. In both cases, the level of reliability places an upper bound on a coding scheme's predictive ability. Unitizing reliability, the degree of agreement regarding identification of the units to be categorized, is needed to ensure that multiple coders of the same interaction will be coding from the same set of units. Unitizing reliability is most typically assessed using Guetzkow's U, which calculates the difference between the number of units identified by an independent coder and the "true" number of units (the average of two coders' estimates) using a simple equation

$$U = (O_1 - O_2)/(O_1 + O_2)$$

where $U * 100$ is the percent of disagreement and O is the observer or coder (Folger et al. 1984; Guetzkow 1950). For example, if coder 1 identified 75 units and coder 2 identified 85 units in the same transcript, Guetzkow's $U = (85 - 75)/(85 + 75) = .062$, meaning that there was a 6% discrepancy between the number of units identified by either coder and the "true" number of units. Interpretive reliability refers to consistency in applying labels to the units and is typically measured across coders. That is, multiple coders independently code the same text, and their assignment of categories is compared for consistency. Thus, consistency of interpretation is assessed in terms of the reliable application of interpretive rules. It is also useful to obtain consistency measures within coders across time to check for drift in application of the coding scheme. The most common and complete global measure is Cohen's Kappa (1960), which determines the level of agreement corrected for agreements due to chance. Cohen presents the following formula to calculate Kappa:

$$\text{Kappa} = (P' - P_c)/(1 - P_c)$$

where P' is the observed percentage agreement among coders and P_c is the proportion of chance agreement. Assuming that each unit to be coded has the same probability of accurate classification, then the probability of chance agreement (P_c) is $1/k$, where k is the number of categories in the coding scheme (Folger et al. 1984). For example, if there are 5 categories and the coders agreed on 75% of the codes assigned ($P' = .75$), then $\text{Kappa} = (.75 - .2)/(1 - .2) = .6875$. Additional details for how to calculate Kappa can be found in Cohen's original article (1960). More detailed discussions of Kappa can be found in Fleiss (1971) and Folger et al. (1984).

Our experience suggests that, to ensure high interpretive reliability, it is best to conduct unitizing and coding in separate passes through the data. Doing both simultaneously may result in a disagreement about what code to apply

because coders have unitized (identified the speaking turns) slightly differently rather than because they are using the codes differently. This might occur because some coders treat back channels (“uh-huh,” “okay,” or “I see”) as substantive breaks in the flow of conversation and so unitize around them whereas others do not. This problem can be avoided by completing the unitizing first, checking for unitizing reliability, and then content coding from the same set of pre-unitized tapes or transcripts.

Interpretive validity

Interpretive validity refers to the degree to which a coding scheme taps into the information it was designed to obtain. Different approaches have been used to validate coding schemes, depending on the goals of the research. Poole and McPhee (1994) identify three types of claims that might be made in an attempt to demonstrate validity of a coding scheme. These claims are linked to the source of interpretation of the actions being studied: 1) observer-privileged: where the researcher seeks to explain interaction from the outside without reference to subjects' perspectives, 2) generalized subject-privileged: focusing on shared meanings for members of a culture, and 3) restricted subject-privileged: with the goal of identifying idiosyncratic meanings for people in a particular group or relationship. As one moves from observer to restricted subject-privileged approaches, more input from participants and more intimate knowledge about the work group (as might be experienced by a group member) is required.

Issues to Resolve before Analyzing your Data

So far, we have provided a “how to” for coding data. Having gotten this far, the obvious question is “what next?” Before one can start analyzing the data, there are still a number of issues that need to be resolved. How this is done will be determined, in part, by the kinds of research questions that one is asking. In this section, we consider three related issues: aggregating data, level of analysis and time segmentation.

Aggregating data

The first issue to consider is whether to focus on individual behaviors or to aggregate over individuals and focus on the dyad or group. Is it theoretically important to know how many times each negotiator in a dyad or group uses a specific tactic (e.g. making a demand) or can the hypotheses be tested by

knowing, overall, the number of times that demands were made in a negotiation? In our research, we have largely focused on behaviors aggregated to the level of the dyad or group. This has enabled us to provide a global description of regularities in negotiators' behaviors and the relationships between the negotiating context, negotiators' behavior, and their outcomes. It allows us to ask whether, in general, integrative outcomes are associated with different kinds of strategies rather than distributive outcomes or impasses. It also enables us to question whether, in general, cooperatively-motivated negotiators behave differently from individualistically-motivated negotiators.

Our experiences suggest that while this level of analysis captures many important aspects of negotiators' behavior there are circumstances under which an individual level analysis is important, for example when negotiators differ in terms of power (Giebels, DeDreu, and Van de Vliert 2000) or social motives (Weingart, Brett, and Olekalns 2002). In our research, we have manipulated the composition of negotiating dyads and groups. At the level of the dyad, this means that two negotiators can have similar attributes, for example, both are cooperatively-motivated, or they can have mixed attributes, that is, one negotiator is cooperatively motivated while the other is individualistically motivated. An interesting question is how dyad or group composition affects negotiation processes: Do cooperatively-oriented negotiators behave the same way regardless of the characteristics of the other negotiator or do they adapt their behaviors? To answer questions such as this, it is necessary to shift from analyzing dyadic or group behaviors to individual behaviors within those groups.

Independent of the content of the coding scheme, researchers need to decide how detailed their coding scheme should be. The answer to this question again depends on the level of your theory – whether global or specific behavior is of interest. If your hypotheses are about integrative or distributive behavior generally, you might want to classify comments using those broad categories. We find it is more effective to use more detailed coding categories and then aggregate these categories to the more general level during the analysis phase. This approach makes it less likely that two functionally different behaviors will be subsumed under one code. However, the more detailed the coding scheme the more likely we are to see coding errors and low reliability. Increasing the number of categories also means that the frequency of each category decreases, resulting in sparse data sets that create problems for most statistical methods.

There are several approaches for aggregating tactics into more general groupings. You may choose to adopt a theory-driven approach, aggregating behaviors that theories classify as integrative or distributive. A pitfall for this kind of approach is that it neglects actual usage and so may group behaviors

that are not associated with each other in actual negotiations. Empirical analysis thus provides an important test of our theoretical assumptions about the function and use of specific tactics. For example, we have found that the use of positional information (i.e., information about what outcome a negotiator desires on a given issue) clusters with a focus on offers and concessions, whereas argumentation clusters with threats and demands (Olekalns & Smith, 2003). In a more theoretically-driven approach, we likely would have combined positional information and argumentation into one category (distributive information exchange) and offers, demands, and concessions into a second category (offer management).

To empirically determine clusters of tactics, we use correspondence analysis (Greenacre 1993). This analysis establishes relationships between tactics, based on their frequency of use. It is conceptually similar to factor analysis, but developed specifically for use with count (frequency) data. Figure 1 shows the output from a correspondence analysis of Towers II data (the mapping of data points in the two-dimensional space) with six clusters identified and interpreted. There are no hard and fast rules for the interpretation of this output. Visual inspection determines clusters based on (a) physical proximity and (b) thematic fit. Some categories were mapped in unexpected places. We had to go back to the transcripts and consider what negotiators were actually doing to help us decide how to cluster those behaviors. For example, in Figure 1, asking questions about priorities (category 9), and references to power (category 20) were mapped near one another but negotiation theory would suggest the former is an integrative behavior and the latter a distributive behavior. Rather than put these two categories of behavior in the same cluster, we looked for other proximal behavior that might cluster with each theoretically. As a result we were able to identify theoretically meaningful *and* proximal clusters of negotiation behavior.

Levels of analysis

The second issue concerns the level of analysis. In analyzing your data, you can frame your hypotheses in terms of the frequency with which strategies are used, how they are sequenced or how strategies evolve over time. Each of these levels of analysis answers slightly different questions. In this section, we provide a brief overview of the issues associated with each level of analysis (see Brett, Weingart and Olekalns, 2004; Weingart & Olekalns, 2004). Our other article, to be published in the next issue of *International Negotiation* (10-1), considers the modeling of sequences in greater detail (Smith, Olekalns and Weingart). These three approaches should not be considered as mutually

exclusive. Rather, each provides different kinds of information about negotiation processes and helps build a more complete picture of what negotiators do (e.g., Olekalns & Smith 2000).

Looking at the *frequency* with which negotiators use specific strategies is by far the most common approach to analyzing negotiators' interactions. Using this approach, one simply counts the number of times that a given strategy (for example, a threat), is used. Focusing on the frequency with which strategies are used tells us about the global approach used by the negotiator(s). The major shortcoming of this approach is that it reflects a static view of negotiation processes: counting the number of times that types of strategies occur does not tell us about how negotiations unfold over time. Therefore, while we can draw very broad conclusions about the relationship between external factors and strategy choices, as well as between strategy choices and outcomes, we cannot learn about the patterning of strategies over time. Analyzing frequencies helps us to answer questions such as: Does the negotiating context affect the kind of strategies that are used? Are specific strategies linked to different kinds of outcomes?

There are two approaches that allow us to better capture the dynamic nature of negotiations. The first focuses on how one negotiator responds to another, that is, on how strategies are *sequenced*. Studying sequences allows us to examine patterns in negotiators' interactions at two levels: sequence structure and sequence content. For example, we ask whether negotiators match (reciprocate) or mismatch (fail to reciprocate) each other's behaviors. This tells us whether negotiators are attempting to sustain or redirect the prevailing strategic orientation. A second layer of information is provided by examining the content of these sequences. Analyzing content tells us about the causes and consequences of matching, and helps us to understand how negotiators break off distributive spirals or derail integrative processes. This approach also helps us to develop more complex models of negotiators' interactions: It provides insights into how negotiators constrain each other's actions and into how they direct the negotiation towards obtaining their preferred outcomes. One important insight that we have gained from this type of analysis is that sequences that are influential in determining the outcome of negotiations often incorporate strategies that are not frequently used in their own right. This level of analysis thus complements rather than duplicates frequency level analyses. Our paper on Markov chain analyses provides an introduction to analyzing strategy sequences (Smith, Olekahns, and Weingart, forthcoming).

Although sequences incorporate a temporal dimension, the focus on immediate actions and reactions means that the analysis is of short-term effects. Analyzing negotiation *phases* also captures the temporal aspect of negotia-

tions. However, the time frame for examining the consequences of strategies is a longer-term one, allowing consideration of the ebb and flow of the negotiation over time. There are two approaches to analyzing negotiation phases. The first, a *stage* model approach, treats negotiations as divisible into discrete time segments and considers how the frequency of different strategies changes across segments. The second approach, called an *episodic* approach, looks for naturally occurring phases: Phases are clearly identifiable interaction sequences with explicit beginnings and ends (Baxter 1982). They are defined on the basis of 'runs' of strategies (i.e., the continuous use of a single strategy) with transitions between phases identified on the basis of interruptions to extended runs of the same strategy. In the following section we consider the methodological and empirical issues associated with adopting either of these approaches. Comparing sequence and phase analyses can highlight negotiations in which these temporal aspects of negotiation work together and reinforce each other and those in which they work in opposition.

Time segmentation

Researchers who want to examine negotiation phases are faced with a choice between a stage model and an episodic model. The principal advantage of a stage approach is that it ensures comparability across research; its principal disadvantage is that, by arbitrarily determining when stages begin and end, it is likely to miss naturally occurring shifts in the negotiation process. If one chooses to work within a stage approach, one still needs to make several decisions. The first is how many stages in which to divide the negotiation. Negotiation theory identifies anywhere between three and 12 stages. In our research, we have used six stages as the number of segments that provides a reasonably fine-grained analysis of negotiation while maintaining a reasonable number of observations within each segment (e.g., Olekalns, Smith and Walsh 1996). The second is how to determine these stages. You can do this either on the basis of time (divide each negotiation into, for example, five minute intervals) or on the basis of speaking turns (for example, if the negotiation is made up of 300 speaking turns, then each 50 speaking turns become a stage). The principal disadvantage of a time-based segmentation is that negotiations do not all take the same amount of time. You may thus find yourself comparing strategies that occur in the 10–15 minute interval and which represent the midpoint for one negotiating dyad but the endgame for another. We therefore use segmentation based on speaking turns. We prefer this approach as it means that we are comparing comparable segments of the negotiation: midpoint to midpoint and endgame to endgame.

Conclusion

In this article, we have provided an overview of the decisions that researchers need to make when they embark on an analysis of negotiation processes. Whether you choose to use an existing coding scheme or to develop one that meets your needs, you will need to make decisions about the level at which to code data, whether to aggregate tactics into broader strategies and on the level at which to analyze your data. You will need to decide whether to focus on what strategies negotiators use, how strategies combine to facilitate or inhibit goal attainment, or how negotiations move from beginning to end. We have found that each level of analysis brings with it unique insights into the negotiation process. We would be less than honest if we did not conclude by saying that this is a labor-intensive and time-consuming process. However, we believe that those of you who persevere will find the richness of your data to be its own reward.

References

- Bales, R.F. (1950). *Interaction process analysis: A method for the study of small groups*. Cambridge, MA: Addison-Wesley.
- Baxter, L.A. (1992). "Conflict management: An episodic approach," *Small Group Behavior* 13:23–42.
- Brett, J.M., Weingart, L.R., and Olekalns, M. (2004). "Baubles, Bangles and Beads: Modeling the Evolution of Negotiating Groups over Time." In M.A. Neale, E.A. Mannix, and S. Blount-Lyons (Eds.) *Research in Managing Groups and Teams: Time in Groups (Volume 6, pp. 39-64)*. New York: Elsevier Science.
- Carnevale, P.J.D., Pruitt, D.G., and Seilheimer, S.D. (1981). "Looking and competing: Accountability and visual access in integrative bargaining," *Journal of Personality and Social Psychology* 40:111–120.
- Cohen, Jacob (1960). "A coefficient of agreement for nominal scales." *Educational and Psychological Measurement*, 20, 37–46.
- Donohue, W.A., Diez, M.E., and Hamilton, M. (1984). "Coding naturalistic negotiation interaction," *Human Communication Research* 10:403–425.
- Fleiss, J.L. (1971). "Measuring nominal scale agreement among many raters," *Psychological Bulletin* 76:378–382.
- Folger, J.P., Hewes, D.E., and Poole, M.S. (1984). "Coding social interaction." In B. Dervin and M. Voight (Eds.) *Progress in communication sciences (Vol. 4, pp. 115–161)*. Norwood: Ablex.
- Giebels, E., De Dreu, C.K.W., and Van de Vliert, E. (2000). "Interdependence in negotiation: Effects of exit options and social motive on distributive and integrative negotiation," *European Journal of Social Psychology* 30:255–272.
- Greenacre, M.J. (1993). *Correspondence analysis in practice*. London: Academic Press.
- Guetzkow, H. (1950). "Unitizing and categorizing problems in coding qualitative data," *Journal of Clinical Psychology* 6:47–58.

- Hewes, D.E. (1979). "The sequential analysis of social interaction," *Quarterly Journal of Speech* 65:56-73.
- Hyder, E.B., Prietula, M.J., and Weingart, L.R. (2000). "Getting to best: Efficiency versus optimality in negotiation," *Cognitive Science* 24(2):169-204.
- Lewicki, R.J., Saunders, D.M., and Minton, J.W. (1985). *Negotiation*. Irwin: Boston.
- Olekalns, M., and Smith, P.L. (1999). "Social value orientations and strategy choices in competitive negotiations," *Personality and Social Psychology Bulletin* 25:657-668.
- (2000). "Negotiating optimal outcomes: The role of strategic sequences in competitive negotiations," *Human Communication Research* 24:528-560.
- (2003). "Testing the relationships among negotiators' motivational orientations, strategy choices, and outcomes." *Journal of Experimental Social Psychology*.
- Olekalns, M., Smith, P.L., and Walsh, T. (1996). "The process of negotiating: Strategies, timing and outcomes," *Organizational Behavior and Human Decision Processes* 67:61-77.
- Poole, M.S., Folger, J.P., and Hewes, D.E. (1987). "Analyzing interpersonal interaction." In M.E. Roloff and G.R. Miller (Eds.) *Interpersonal processes: New directions in communication research* (pp. 220-256). Newbury Park, CA: Sage.
- Poole, M.S., and McPhee, R.D. (1994). "Methodology in interpersonal communication." In M. Knapp and G.R. Miller (Eds.) *Handbook of interpersonal communication*. Newbury Park, CA: Sage.
- Pruitt, D.G. (1981). *Negotiation behavior*. Academic Press: New York.
- Pruitt, D.G., and Carnevale, P.J. (1993). *Negotiation in social conflict*. Pacific Grove: Brooks/Cole.
- Pruitt, D.G., and Lewis, S.A. (1975). "Development of integrative solutions in bilateral negotiation," *Journal of Personality and Social Psychology* 31:621-633.
- Putnam, L.L., and Jones, T.S. (1982). "The role of communication in bargaining," *Communication Monographs* 3:262-282.
- Smith, P.L., Olekalns, M., Weingart, L.R. (in press). Markov chain Models of communication processes in negotiation. *International Negotiation*
- Trujillo, N. (1986). "Toward a taxonomy of small group interaction coding systems," *Small Group Behavior* 17:371-394.
- Walton, R.E., and McKersie, R.B. (1965). *A behavioral theory of labor negotiations*. McGraw-Hill: Boston.
- Weingart, L.R. (1997). "How did they do that? The ways and means of studying group processes." In L.L. Cummings and B.M. Staw (Eds.) *Research in organizational behavior* (Vol. 19 pp. 184-239). Greenwich: JAI.
- Weingart, L.R., Bennett, R.J., and Brett, J.M. (1993). The impact of consideration of issues and motivational orientation on group negotiation process and outcome. *Journal of Applied Psychology* 78(3):504-517.
- Weingart, L.R., Brett, J.M., and Olekalns, M. (2002). *Conflicting social motives in negotiating groups*. Paper presented at the 62nd Annual meeting of the Academy of Management, Denver, Colorado.
- Weingart, L.R., Hyder, E.B., and Prietula, M.J. (1996). "Knowledge matters: The effect of tactical descriptions on negotiation behavior and outcome," *Journal of Personality and Social Psychology* 70:1205-1217.
- Weingart, L.R., and Olekalns, M. (2004). "Communication Processes in Conflict and Negotiation." In M. Gelfand and J. Brett (Eds.) *The Handbook of Negotiation and Culture: Research Perspectives*. Palo Alto CA: Stanford University Press.
- Weingart, L.R., Thompson, L.L., Bazerman, M.H., and Carroll, J.S. (1990). "Tactical behavior and negotiation outcomes," *The International Journal of Conflict Management* 1:7-31.

Copyright of International Negotiation is the property of Martinus Nijhoff and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.